

TUTORIAL 12: 16.5

Find (a) the curl and (b) the divergence of the vector field.

$$(5) \quad F(x, y, z) = \frac{\sqrt{x}}{1+z}i + \frac{\sqrt{y}}{1+x}j + \frac{\sqrt{z}}{1+y}k.$$

$$(8) \quad F(x, y, z) = (\arctan(xy), \arctan(yz), \arctan(zx)).$$

(14) Determine whether or not the vector field is conservative. If it is conservative, find a function f such that $F = \nabla f$. $F(x, y, z) = xyz^4i + x^2z^4j + 4x^2yz^3k$.

(28) Prove that $\operatorname{div}(\nabla f \times \nabla g) = 0$ assuming that the appropriate partial derivatives exist and continuous.